

Industries, universities and the technoscientific complexes

LET'S FIRST SURVEY THE PRESENT relations between universities and commercial companies in high-tech industries. From the universities, the companies receive graduate scientists for employment at various levels, ranging from first-degree holders through trained researchers to leading experts capable of establishing major new industrial research programmes in the company's laboratories. They may well also receive formal or informal advice from academic consultants who draw on personal knowledge and experience, or perhaps on the work of their university research groups. The company library contains scientific periodicals, the contents of which are mostly written by academics. Perhaps the company researchers may use university facilities from time to time, though for large companies the relation may now be reversed. From the companies the academics receive research support – perhaps grants for postgraduates to work on topics of interest to the company, perhaps a contribution to the running costs of the laboratory. Or the support may be in kind – the supply of instruments or of research materials such as rare chemicals or specially bred laboratory animals. In addition to direct support from companies, academics may benefit from government or university funds because the research seems likely to be important for industry and their students work on problems at the cutting edge of technology.

The interchange of personnel, information and resources is now very considerable; few high-tech firms and few leading departments could manage without it. But as I mentioned earlier,

the networks also include other organisations: for medical sciences and technologies, hospitals are crucial, but general practitioners may also be involved – in clinical trials and more generally as purchasers of medical products. In fields with military importance, scientists and engineers employed by the armed forces are central. The extensions to agriculture are obvious. In all these fields, government laboratories may be involved as innovators; more generally they provide standards, test materials, check specifications, assess needs, etc. In the most expensive fields of science, where equipment costs more than universities or companies can afford, then governments may be persuaded to oblige – individually or collectively. Academics may thus gain access to rare facilities, and industrialists gain contracts to construct the equipment and benefit from ‘spin-off’ technologies. Governments gain prestige, and hope for military or industrial advantage.¹

These are the *technoscientific* complexes which dominate the knowledge and commodity production of our time. For our analysis of them, as I suggested earlier, we gain more by recognising the dense interweavings of universities and industry, and of science and technology, than we do by trying to separate them. This chapter is about the development of the interactions, about the objects, knowledges and skills that moved between industry and academies, and about the social relations that were both the prior conditions and the products of those movements (Latour, e.g. 1987a; 1987b).

In the last chapter I introduced ‘synthetic inventions’ as the analogues of ‘synthetic experiments’ and suggested that these two kinds of synthesis were sometimes intimately related. Typically, the ‘model systems’ of laboratories could provide either light or fruit, to use Bacon’s terms; where both philosophical and practical aspects were pursued, such models become the core of *technosciences* – of institutions and projects that produced knowledge-commodities. If we see technoscience as a way of *knowing* and a way of *making*, we underline the confluence of the scientific and the technological histories with which this chapter is concerned. We also allow for *different kinds* of technoscience. For example, major expeditions could be thought of as

natural-historical technoscience – networks of academics, government and industry in pursuit of specimens. The networks of professional institutions and government/military services in post-Revolutionary France might usefully be called *analytical technoscience*, and early in this chapter we will explore further some of the analytical relations between academics and industry in the nineteenth century. But this chapter goes on to trace the closer forms of academic–industrial collaborations that I have called synthetic technoscience – when academics and industrialists work on model systems that are philosophically *and* commercially interesting, and when synthetic experiments/inventions are developed in networks of universities, research institutions and industrial research laboratories.

I suggested previously that such networks involved all our ways of knowing and of making – natural history and craft, analysis and rationalised production, and synthetic experiments and inventions – but we can also note that they could be variously ‘directed’. Technoscience networks were not all ‘industrial’, some projects were ‘governmental’ in as much as they were led by state agencies for military, public health or national prestige reasons. Some other projects were directed by academic communities which had persuaded governments that the likely products or prestige merited public investment.

This chapter surveys the key industrial technosciences of the late nineteenth century (electrotechnics and pharmaceuticals) and then outlines some of the key state projects in the two world wars (e.g. poison gas and the atom project). The state-academic-industry linkages built in the First World War were often transitory, those of the Second World War continued to be developed post-war, in the USA and the UK. I conclude by surveying the relative decline of state-technoscience and the continued rise of industrial-academic linkages at the end of the twentieth century.

But in tracing the developing intimacy of science and technology, we must always remember that for most of the nineteenth century, and for many sectors up to the Second World War, the development of industry was not heavily dependent on universities or other forms of higher scientific education. And where it

was so dependent – for chemistry in most countries, and in Germany for most technologies – the symbiosis was largely at the level of *analysis*.

Analysis and established technologies

In Chapter 4 we have already discussed some of the characteristic nineteenth-century interactions that we might call *analytical* technoscience, including the use of graduates (especially chemists) as consultants, and their emergent roles in public regulation of health and hazard. Especially in the chemical industries, many proprietors had a good knowledge of science and good connections with local lecturers and scientific societies. In mechanical and civil engineering the same networks operated – between proprietors, consultants and teachers – mostly to solve practical problems. Eaton Hodgkinson serves as a Manchester example; a ‘devotee’ mathematician who worked with William Fairbairn on design of girders, and on a radical new ‘tubular’ bridge over the Menai Straits, he later became professor of engineering at University College, London. Hodgkinson conducted experiments on strength of materials, but these were done in Fairbairn’s works (Rosenberg and Vincenti, 1978). In many areas of engineering, the technical problems were solved by trial and error, or by systematic testing rather than by calculation. Joseph Whitworth, for example, developed his rifles on a testing-ground at his home in the suburbs of Manchester (Musson, 1975).

Few of the ‘great (British) engineers’ were graduates, but that did not mean they relied only on craft skills. Many of them had been apprenticed in large engineering works – which remained a preferred means of training until the end of the nineteenth century. Most of them understood some mathematics, read the engineering magazines, pursued patents and knew about developments elsewhere; they were often major sponsors of formal scientific education – thus enlarging both the community of experts and the supply of educated workers. The mechanics’ institutes that made science available to the working classes in

most industrial towns after *c.*1825 usually numbered engineers among their supporters, and the science colleges and provincial university colleges founded from the 1850s were similarly supported (Inkster, 1997; Inkster and Morrell, 1981).

Owens College, Manchester, after an uncertain start, flourished from the 1860s in part because its chemistry professor, Henry Roscoe, could count on the support of key local industrialists. A chair of engineering was created and filled by Osborne Reynolds, chiefly through the support of local engineers; by the end of the century, a university degree was a suitable prelude to a high-level apprenticeship. Joseph Whitworth gave part of his fortune to establish scholarships in engineering, and much of the rest went to the university, the technical college and the new teaching hospitals. In turn, the university professors and their graduates acted as advisers, consultants or managers for local industry (Kargon, 1977).

That was the English pattern, but Germany from the 1830s was the chief foreign model in such matters, and one that was sometimes imported. The leading chemists of mid-century Britain had mostly trained in Germany, until Hofmann (whose atomic models we discussed) extended that educational facility to London. Some German chemists worked in British industry temporarily, e.g. Heinrich Caro who returned from Manchester to lead German dyestuffs research; some stayed in Britain and built chemical businesses, often becoming advocates of technical education.

In Germany, the higher technical colleges taught engineering (Lundgren, 1990). Many of the leading engineers were educated in them and some of the graduates moved to Britain or the USA, where they pressed for the inclusion of engineering in universities. Several of the engineers in Manchester, e.g. C. F. Beyer in locomotive manufacture, and Henry Simon in flour milling, had trained in German (or Swiss) polytechnics, and were major supporters of the university and technical college (Simon, 1997). But even in 'advanced' cities like late nineteenth-century Manchester, with its strong community of science teachers and education-minded industrialists, each side knew its place.

Chemists in the new university taught analysis and developed experimental disciplines; they acted as advisers to industry, but did not themselves experiment with industrial processes. That closer interaction *might* have occurred in technical schools, in the programmes of education for specific trades which began to develop from the 1850s, but there was little state support until the 1890s, and problems of resources and of industrial secrecy and rivalry prevented much close engagement with specific technologies.

The situation was a little different in the other great capital of industrial Britain – Glasgow. As a Scottish city, Glasgow had a much stronger tradition of public education and a long-established university, including a medical school. Glasgow, as we noted, had a chair of natural philosophy to which they attracted William Thomson, an accomplished mathematician and experimentalist, much interested in engineering problems, not least the emergent electrotechnics. But Glasgow from 1838 also had a chair of engineering, the holder of which made major contributions to the analysis of steam engines (Channell, 1988). This formal support for activities which were informal in English regional capitals encouraged closer relations between academe and industry, especially around telegraphy.

By c.1900 the leading universities and colleges of Britain had laboratories where students could ‘practise’, and in the twentieth century most consultant engineers had formal training, at least in evening classes in technical schools (Guagnini, 1991); but we should not exaggerate the impact of higher education, or systematic research, on industry generally. Many forms of manufacture that were new(ish) c.1900, e.g. rifles, bicycles and cars, involved lots of invention but few inputs from academics. British motor manufacturers conducted little ‘research’ and employed few ‘graduates’ until after the Second World War (Buchanan, 1989). ‘Developments’ were usually combinations from ‘prior arts’; relatively little came from ‘laboratories’. Certainly that is true compared to ‘electrotechnics’, which became one of the key synthetic technosciences of the later nineteenth century.

Electrical analysis and synthesis

In some senses electrical technologies were born in the laboratories of natural philosophers; they were 'born into' worlds of measurement, quantification, demonstration and the search for principles.² The relation between electrical principle and practice was close, perhaps especially in Britain where the key discoveries of the early nineteenth century coincided with strong amateur cultures of science and technology. Faraday at the Royal Institution was the archetypal electrical experimentalist, and his audiences contained many professionals and manufacturers with practical interests (Berman, 1978). A taste for natural philosophy was well developed in middle-class and working-class circles, especially in the industrial towns; and savants, artisans and inventors concerned with steam engines might also be interested in batteries, motors and dynamos.

Faraday himself did not pursue phenomena for commercial advantage (though he spent much time on practical matters such as mine safety and lighthouses), but his close associates were ready to take up his experiments and develop their uses. When Faraday showed how moving wires within a magnetic field could generate an electric current, his friend, Charles Wheatstone, developed the effect to create the electric telegraph. (The American inventor of the telegraph, and the code, was Samuel Morse.) Wheatstone was primed to a 'need' by previous work on the transmission of sound. He worked in his family company making (innovative) musical instruments, and he recognised electricity's potential as a better way of transmitting vibrations (Dostrovsky, 1976). For the most part, telegraphy continued to be developed by inventors and engineers but, as in the case of tubular bridges, difficulties were sometimes met by resorting to systematic testing and perhaps by more sophisticated forms of analysis.

As technologies were extended, so esoteric, analytical advice might prove more necessary. Creating a transatlantic cable telegraph proved to be a major technical challenge that eventually outstripped the know-how of the engineer to whom the company

had entrusted the technical direction; but William Thomson (later Lord Kelvin) had predicted from the beginning that very small currents would have to be used, requiring very sensitive detectors such as he had designed. He was proved correct and his Glasgow university laboratory proved invaluable – for instrumentation, for testing cable materials and for training telegraph employees; student laboratory exercises included real industrial tasks; part of a university department became interdigitated with engineering companies. Thomson's science rescued a very expensive, high-prestige project and it made him a fortune (Smith and Wise, 1989).

Telegraphy was a major new communications technology of great commercial and political importance, especially for imperial powers, but its impact was far surpassed by the development of electrical supply systems in the last two decades of the century, first in the USA. These were initially for electric lighting, but could also be used for electric power systems for tramways (Hughes, 1983). One of the consequences was a huge demand for electrical engineers and especially technicians. The development of technical education at the end of the century centred on this demand (which was also met by private courses). In a sense then, it was for electrical technology that formal education became routine for technicians, as it was for telegraphy that testing laboratories had become routine for engineers.

One can see some of the consequences at university level by looking again to Manchester. In 1900 a new *institute* was opened for physics and 'electrotechnics'. It contained a large laboratory filled with the latest electrical equipment – generators, motors, lighting systems, etc., on which the students could 'work'. The laboratory was closely associated with major electrical firms in the region, and it was named after John Hopkinson, an academic-engineer from a Manchester family of engineers and academics, who had been killed in a climbing accident when aged 49.

John Hopkinson had trained in science at Manchester and in mathematics at Cambridge before relinquishing his fellowship at Trinity College to work for an optical company in the West Midlands. He became interested in electricity through the influence of Thomson. After six years he moved to London as a

consulting engineer and remained in practice after appointment to a chair in engineering at King's College, London, where he directed a laboratory sponsored by the Siemens Company. There he worked on alternating currents, especially the design of dynamos, on which he also collaborated with his brother who, again after an elaborate education at Manchester and Cambridge, had become head of the new electrical division of Mather and Platt, a large Manchester engineering firm best known for textile machinery. John also served on committees that established standards for electrical lighting (Dibner, 1972; Fox and Guagnini, 1999: ch. 3; Guagnini, 1991). Such careers nicely demonstrate how university training, advice and research became important to the electrical industry, but the synergy can also be seen from the 'other side', through developments within industry, especially of research laboratories.

There our best guides are studies of American firms, whose early and successful research laboratories have attracted several good historians (Hounshell and Smith, 1988; Israel, 1992; Reich, 1985; Wise, 1985; and see Dennis, 1987). Such laboratories are easily seen as the hearts of modern industries, as the epitome of technoscience, but we need to be careful about dates, and about the variety of industry. Whatever the strength and extent of that model around the year 2000, it was rare in 1900. I have outlined the British history first, partly because we need to be wary of seeing science and technology as inevitably converging into the industrial research laboratories pioneered by American electrical firms and German chemical companies. While *some* such 'convergence' might be seen as over-determined in the long-term – as secured sooner or later in most of the relevant sites – the form and dating of developments was highly contingent, and the pattern was not widely established until after the First World War. Indeed, *several* notes of caution are in order against the common conflation of industrial laboratories with applied science and with the development of new products (and the equation of any of these with economic success) (Edgerton, 1996b).

First, some of the early industrial laboratories, such as that

established by Thomas Edison in 1870, might best be seen as factories for invention; science was marginal. Inventors had come to the fore because competition between telegraph companies had created hopes that technical improvements would secure relative advantage. Edison established a laboratory because he had gained a contract and recognised that he needed a lot of new devices, and fast. Particular industrial conditions then favoured rapid (synthetic) invention, but that was not the only specific contextual factor. American firms developed industrial laboratories c.1900 in part because of anti-trust legislation which made it difficult for them to buy up firms which held patents they wished to acquire (or block). In Germany, industrial laboratories had developed only after 1876, when changes in patent law prevented the customary 'copying' of new dyestuffs. Such features of the 'regulatory environment' are always strong determinants of technoscientific patterns.

Secondly, much industrial research and development continued to take place outside 'laboratories'. Inventors did not disappear, and analyses could be conducted and experiments mounted in factories or in the field. Indeed, as Fox and Guagnini have claimed, the generating station that Ferranti built at Deptford to supply power to London could reasonably be regarded as one large, complex experiment. It cost a fortune, the technologies were untested on that scale, and it took years to get into working order. Some experiments would not fit into laboratories! (That is one reason for analysing STM in terms of its *projects*, rather than giving primacy to its *places*.)

Thirdly, we need to note that laboratories were not necessarily for the creation of new products; most company laboratories were established to carry out analyses for the better control of production, and that remained the major function of most of them. They might be used to investigate modifications of processes or materials, and this task might approximate 'invention', but *very* few laboratories before 1900 focused on novelty. This is *not* a sign of 'backwardness' – most industrial advance was (and is?) by incremental improvements achieved 'on site' or through analytical/testing laboratories.³ And, of course, even where 'synthetic' work

was central, analytical functions remained important, as we saw in the last chapter for experimentalist laboratories.

In sum, industrial laboratories were important for many forms of analysis, of product and of processes, and they became important in some industries for invention/synthesis and 'development'; but not all of these activities were to be found in any given laboratory, and for none of them was a laboratory a necessary condition. *And yet*, in some companies, the industrial laboratory became the site where all these activities could be brought together, where all the relevant ways of knowing and making could be mobilised and linked with universities and government agencies. The sum of those relations was industrial, synthetic technoscience.

Electrotechnics and industrial laboratories

As noted, the key American exemplar of systematised invention was Thomas Edison, a self-educated telegraph operator turned inventor, who knew little mathematics but was widely read in practical science. In 1870 he won a contract to improve the New York stock-ticker system, and to that end he set up a private laboratory with fifty employees. The laboratory, expanded in 1876 at Meno Park and in 1887 at West Orange, was his tool for systematic invention. By making a business of inventing, Edison managed to put together all the elements of an electrical supply system. By the late 1870s he was seen as the chief author of a technology set to transform the world; he was the hero of the 1881 exhibition in Paris which launched electrical lighting in Europe (Fox and Guagnini, 1998: 125).

His later facilities included extensive libraries (for information) and testing facilities (mostly for analysis), and at their heart was a laboratory for invention where new combinations could be tried. Edison 'invented by design' to fill 'gaps in the market', his methods were based on 'cut and try'; but by the early twentieth century, in his laboratories and in those of his imitators, professional training in science became more important – to

exploit new physical discoveries, to keep better contacts with academics and to recruit talented staff (who were more likely to be graduates as the universities expanded).⁴

An excellent study of the General Electric Company laboratories founded at Schenectady in 1901 reveals the closeness of academic science to certain new industrial sites. For over forty years one of their key investigators was Irving Langmuir, who had graduated in practical science (metallurgical engineering) at Columbia and then done a PhD in Göttingen with Walter Nernst, studying the dissociation of various gases around a glowing platinum wire. Nernst was a world leader in thermodynamics, but also interested in practical problems; he devised a new type of electric lamp that proved very successful. At GEC, Langmuir worked on many key areas of physics, including surface phenomena; he gained a high reputation and a Nobel Prize for his 'science'; but to regard his industrial work as applied science is probably an 'academicist' distortion. It seems to better fit the image pursued in this book – the creation of 'model set-ups' that could be exploited for various purposes, *both* for principles and for practice.

Langmuir began at GEC by trying to increase the longevity of light bulbs, studying various kinds of filament and discovering that residual gases adhered to the glass of the evacuated bulb. The experimental set-up was much the same as Nernst had used for investigating gaseous equilibria around a hot wire. Langmuir was led to investigate the dissociation of hydrogen molecules into atoms; he was also led to propose that bulbs be filled with inert gases rather than evacuated – a method of huge commercial importance. Further studies on the dissociation of hydrogen led to the invention of the atomic hydrogen welding-torch (Reich, 1985; Süsskind, 1973).

One could tell similar stories for industrial research in inter-war England, not least for Metro-Vickers in Manchester, then famed for its industrial scholarships and for laboratories in close relation with the best academic facilities. But first we turn to the other field of early and dense interaction – the dyestuffs-pharmaceuticals complex.

Dyestuffs and pharmaceuticals

The key, again, is synthesis – of new molecules, a few of which had commercial potential. The special conditions included German chemical companies searching for new lines of business, an abundant supply of chemical talent (from university courses which had been boosted for analysis – in pharmacy, medicine and agriculture, as well as industry) and, as we noted, new patent laws in the 1870s which protected discoveries of processes as well as of products. To these we might add the general conditions of the dyestuffs and pharmaceutical industries – the perceived need for new markets, and a premium on novelty arising in part from ‘fashions’. Think of the vogues for new colours for dresses, or the enormous expenditure on patent medicines. Note also the discovery of bacteria from the 1870s and the chance discovery of immunisation – both opening new approaches to therapy for companies that had the scientific and medical contacts.

But let us not anticipate. When synthetic dyes were first discovered it was by accident, by W. H. Perkin, a protégé of Hofmann at the Royal School of Chemistry. His discovery became famous, but it did not lead directly to a new industry because there seemed to be few other such dyes to exploit, and dyers were already producing novelties by traditional craft methods. However, by the later 1870s, several new classes of dyestuffs had been discovered and several German companies were collaborating with university chemists as consultants and to supply graduates. The companies had begun to send graduate chemists to work in university laboratories when they wanted to develop new products. The chemists ‘at the works’ were then largely engaged in analytical work – on the firms’ own products and processes, and on other firms’ products that they might wish to copy, or at least to understand. But the new patent laws placed a premium on ‘home-grown’ products, and the recent discovery of azo-dyes opened new pathways. Some firms ‘pulled back’ their employees from collaborating universities, and concentrated them in industrial laboratories designed to develop new products – backed by analytical and informational services. Invention

here, as in Edison's facility, became 'mass-work'. Lots of chemists tested lots of compounds – few succeeded – but in a market built on fashion, one good novelty was worth a lot of money (Beer, 1959; Fox and Guagnini, 1998: ch. 1; Travis, 1992).⁵

By the end of the century the pattern was well established and reacting back on the educational system. Technical directors of major firms pushed for more practical courses in universities and colleges. When they failed, they supported new ventures in the new century, including a chemical institute under the auspices of the Kaiser Wilhelm Society, an umbrella for institutions meant to boost German industry by encouraging closer relations between principled and practical aspects of research and teaching – such as seemed to be developing in the USA.

The methods for dyestuffs were also used for new pharmaceuticals (or new versions of old remedies), which included over-the-counter remedies such as aspirin, most of which were palliatives to diminish symptoms. However, as industrial chemical laboratories were establishing themselves, medical science was undergoing something of a revolution – bacteria were being isolated for most of the common epidemic diseases – a new form of medical analysis that isolated *causal* elements, rather than tissues or cells that were affected by disease. The causes could be seen under microscopes, at least when dyestuffs were used to colour them (Lenoir, 1988).

Here was a nice connection. Dyestuffs attached themselves to fibres from the cotton plant and/or hairs from sheep – that was their *raison d'être* – so, they could also be used by investigators looking through microscopes at the tissues and cells of animals and plants. Natural dyes such as haematoxylin were part of the tool kit of the microscopists. Bacteria were simple plants, studied by microscopists; they, too, could be stained, indeed their staining properties helped in their classification. Just why particular stains attached to some biological materials and not others was unclear, but the fact was evident, and in the age of bacteria it attained fresh significance. Could one find a dye or other chemical that would attach to bacteria and kill them, without killing

the cells of the animal or person who hosted the bacteria? Hence the search for the 'magic bullet' as envisaged by Paul Ehrlich, whose story is worth telling – for in such lives the elements could mingle without characterisation as either 'science' or 'technology'; and by such lives the technoscientific networks were created.

Remedies for/from microbes

From his student days in the mid-1870s, Ehrlich was fascinated by microscopy, especially the use of new dyes as stains for microscopic specimens. As a medical student he pursued these interests, surrounded by several key creators of the new 'bacteriology'. As a clinician he continued research on the selective affinities of dyestuffs, blood cells and bacteria. One of his dyestuffs, methylene blue, stained nerve fibres, and he showed it could also act as a painkiller. When in 1891 he discovered a stain for the newly discovered parasites in malaria, he administered the dye to two malarial patients, with some success. After a marriage to a textile heiress (note the synergy) and a bout of tuberculosis (note the 'incentive system'), he set up a private laboratory to investigate immunity – another kind of specificity, first uncovered by Pasteur's work on vaccines.

Pasteur had found by chance that weakened bacteria could induce resistance to disease (as cowpox inoculation had been known, since 1800, to produce resistance to smallpox). Pasteur's Institute soon showed that the immunity was in the blood – that proteins (antibodies) were produced which attached to the bacteria (antigens). Here was another new form of medical analysis which could be pursued in laboratory animals, and which had obvious therapeutic potential. It was Ehrlich who showed that various toxic proteins could also act as antigens and stimulate the production of anti-toxins. His studies, in turn, were connected with those of Robert Koch's associates, who pioneered the use of anti-toxins as treatment for diphtheria – a very nasty disease occurring mostly in babies and small children. Although

not without its critics, then and since, diphtheria anti-toxin has usually been counted as one of the key therapeutic fruits of the new bacteriology (Weindling, 1992).

In 1895 Ehrlich was approached to direct a 'serum station' tasked with investigating and *standardising* anti-toxic sera. Standardisation was a problem, in part because the sera could not be analysed chemically for effectiveness; they had to be assayed in tests on animals. One key role of governments in the new 'biological' pharmaceuticals (vaccines and anti-toxins) was to help standardise them, so doctors knew how much they were giving. As we shall see, that state function has been recurrent in twentieth-century technosciences, and it has often provided a basis for adding some experimental work. Indeed, in 1899 Ehrlich was given a new (state) institution in Frankfurt, where he completed his diphtheria studies and began many new researches on chemotherapy. In Frankfurt he was close to a dyestuffs company, Farbwerke Cassella and Co., with which he had already collaborated. They made compounds to his specifications; he discovered that Trypan Red stained the trypanosome parasite and appeared to cure infected mice. The most famous of his products was Salvarsan, hailed as a cure for another parasite disease – syphilis. It was dangerous and difficult to use, but it seemed to herald a new world of rational therapeutics (Dolman, 1971; Lenoir, 1988).

We note the elements which were combined in Ehrlich's work – academic research, clinical research and practice, state support for new remedies and standardisation, and collaboration with chemical companies. This was the recipe for the new pharmaceuticals, taken up in the USA, and then in Britain and France, chiefly after the First World War (Davenport-Hines and Slinn, 1992; Galambos and Sturchio, 1997; Goodman, 2000; Liebenau, 1987; Weatherall, 1990). By then, the British government had established the Medical Research Council, which promoted research, standardisation and clinical trials (Austoker and Bryder, 1989). At this time, industrial laboratories were characteristic of most leading 'technical' firms in the 'new' chemical, pharmaceutical and electrical industries, even in Britain

(Edgerton, 1996b; Edgerton and Horrocks, 1994; Sanderson, 1972).

Science and industry in and after the First World War

It is perhaps in warfare that we can see the most striking extensions of technoscientific networks in the service of governments. The best known example is the mobilisation of scientists in North America to produce the atom bomb during the Second World War, but first I want to retail a brief case study of the most controversial aspect of science in the First World War, the poison gas programme. I do so partly to illustrate the scale and complexity of the mobilisation, and partly to link with the institutions and themes that we have discussed for industrial technoscience at the end of the nineteenth century. The First World War was the great industrial war (Pick, 1993).

The first use of poison gas was by the Germans in April 1915, when a huge cloud of chlorine was released from 6000 cylinders, to drift across the trenches of the French-Algerian forces. From the beginning of the war, both sides had been considering the use of irritating 'tear' gases; the German government took advice from the leaders of its scientific and industrial communities, including Walter Nernst, whom we mentioned above. Research into the various possibilities was conducted at the Kaiser Wilhelm Institute for Physical Chemistry and Electro-chemistry in Berlin, one of the turn of the century institutes developed by the state in partnership with industry to perform 'useful' research which seemed beyond the capacity or the inclinations of university departments. The chief author of the poison gas programme was Fritz Haber, a well-known chemical technologist, the head of another Kaiser Wilhelm Institute which became part of the military command for the duration of the war (Mendelsohn, 1997).

The USA and the UK also organised investigation and production of poison gases. The USA used its Bureau of Mines to establish a chemical warfare service based at the American

University in Washington, DC. One of the leaders in the work on mustard gas was J. B. Conant, later a major figure in the atom project (and a sponsor of post-Second World War research and teaching in history of science). The production plant for these deadly chemicals was not in Washington, but in the industrial city of Cleveland, Ohio. In Britain, many university laboratories were commandeered; and providing remedies against the gases became a major research programme for the newly established MRC. Here it could draw on previous research in respiratory physiology, performed in connection with the coal-mining companies (Sturdy, 1992a).

As Everett Mendelsohn underlines in his account of poison-gas programmes, few if any of the scientists had any qualms about the morality of chemical warfare. Conant was clear that poisoning soldiers with gases was morally indistinguishable from blowing them apart with explosives. The majority of leading German scientists saw German militarism as the spearhead of Western civilisation, and the allied scientists mobilised against the 'threat of the Hun'. The internationalism which had been increasingly prominent in STM since the 1880s crumbled rapidly; the power of nationalistic, state-directed technoscience was becoming horribly apparent.

In the First World War, much more than in the Second World War, most of the technoscientific programmes were improvised by government and military agents whose prior experience of technical programmes was limited. Famously, one of Ernest Rutherford's brightest assistants in the Manchester laboratory went to his death as an ordinary soldier in the trenches. But the scale of the mobilisation became huge and the British state gradually learned to use its STM talent to better effect. It stopped sending physicists and chemists to fight in the trenches, and used them instead to improve ballistics or to work out techniques for gas warfare.

The medical services, too, came to be highly organised, giving scope to new would-be specialists in cardiology or neurology. A hierarchy of casualty stations and hospitals was established, reaching from the trenches to the cities back home, where hundreds of

public buildings and private homes were taken over as hospitals. At least in principle, this was the very model of medicine as *analysis and rational production*. In former tuberculosis sanatoria and mental asylums, special war hospitals were established in which doctors worked out new kinds of medicine – ‘functional’ approaches, based on experimental physiology and intended to get men back to the front (Cooter, 1993a; 1993b; Sturdy, 1992a). Though most of this huge machine evaporated at the end of the war, the research institutions emerged stronger, and some doctors had acquired a taste for ‘organisation’ that was renewed during the next war in debates about a National Health Service.

The effects of the First World War on British industry were rather similar. When it could no longer import German fine chemicals, pharmaceuticals or optical glass, the British government had to set up research teams and factories to manufacture substitutes – hammering home the lesson about German technical superiority which had been a regular complaint of British scientists since the 1880s. Huge supply networks were established, and researchers were mobilised to improve the workers as well as the products. The munitions factories, where most of the employees were women, served as laboratories for the study of fatigue – physiological medicine on the shop floor (Sturdy, 2000). When the war was over, some of the new organisation remained; the government set up a research council to fund scientific research in universities, and it supported research associations and laboratories for the major industries. In a sense, British industry was spurred to ‘catch up’ with German patterns, and perhaps especially with ‘American’ firms such as Westinghouse, Ford and Burroughs-Wellcome which had become major players in Britain in the years before the war (Alter, 1987; Edgerton, 1996b; Edgerton and Horrocks, 1994).

Historians have paid insufficient attention to the *ending of wars*, to what remained of the wartime structures, to the differences between peace-time institutions ‘before and after’. The long-term effects of the First World War on technoscientific networks differed between countries, but generally they were less than for the war that followed. In Britain and France, the con-

nections made in the First World War helped create a series of state-funded research institutes, for civilian and military programmes. In the USA, most of the military-STM links were abandoned, as part of the distancing from a dreadful conflict. In the 1920s the most conspicuous funders of American public research were the foundations (especially the Rockefeller Foundation and the Carnegie Foundation⁶), rather than the Federal government; the usual vehicle was university departments, which became increasingly oriented to research rather than teaching (Geiger, 1992; 1997; Mendelsohn, 1997). By the 1930s, in most Western nations, there were extensive and intensive bonds between *universities and companies* in chemistry, electrical technology (including electronics) and in the new pharmaceuticals.

In each country, in each such field, a handful of key researchers and consultants moved easily between universities, companies and government agencies. Henry Dale in the UK and A. N. Richards in the USA were 'everywhere' in pharmaceuticals (Swann, 1988). The Cavendish physics laboratory in Cambridge enjoyed close relations with Metro-Vickers's laboratory in Manchester (Crowther, 1974); the company supplied equipment and sought advice, Cambridge supplied prestige as well as information. The laboratory of Imperial Chemical Industries in Cheshire sometimes seemed like an Oxbridge college, and had contacts to match. (By contrast, their dyestuffs laboratory in north Manchester seemed lower-middle class and more obviously 'provincial' in its connections – Reader, 1970, 1975.) And if we look for technoscientific networks in inter-war medical services (rather than the pharmaceutical industry), the clearest example would seem to be radiotherapy for cancer – a small field, but changing rapidly in ways that contrasted strongly with the 'liberal' patterns of most of British medicine. It will serve rather nicely to illustrate the possibilities of technomedicine, even before the Second World War.

Radium, when introduced around 1900 had been the epitome of 'free market' medicine; it was seen as a 'general tonic' and incorporated into patent medicines, albeit in very small quantities. Larger quantities were bought by medical charities and by

public subscription in the hope of 'burning' cancers, but radium was very dangerous and very expensive. From about 1910, central government was increasing its support both for scientific research and for public hospitals. By the 1930s, control of most of the medical radium was vested in the government and the major radiotherapy centres were regulated by the National Radium Commission. The centres were controlled by a new sub-profession of medicine – the radiotherapists – and linked both with physics departments in universities and with companies supplying X-ray generators etc.; they employed 'medical physicists' to help control dose regimens and radiation hazard. The Manchester centre was supported by municipalities across the region because cancer was then emerging as a 'public health' issue. Cancer had already been a field for organised biomedical research since c.1900, often with charity funding (Löwy, 1997); the Manchester research laboratory was attached to the radiotherapy hospital, and was both experimental and analytical. This complex was peculiar in the scale of its work, but it was widely admired as a model of 'scientific organisation' (Pickstone, 1985; Pinell, 2000).

We may conclude that, by 1939, if you had wanted to compare 'new industries' across the different nations, then you needed to include these academic-industrial and medical complexes. Had you wanted to compare *academic provision*, then you needed to take account of governmental support for 'useful' research (which increased rapidly between the wars), as well as industrial support in money and equipment. In your survey of such linkages, you would have noted that *instrument-makers* served both universities and industry, and that *standardisation institutions* were crucial to these technoscientific complexes. So too was empire. In Britain and France, imperial engineering, agriculture and medicine, both governmental and commercial, employed a high proportion of the country's STM graduates, and provided a major field for university researchers (Farley, 1991; Worboys, 1976).

Yet that was not the whole picture. The techno-scientific linkages were often opposed; many academics and doctors resisted

commercial 'encroachment' and even state support. Scientists in industry were excluded from some scientific and professional societies, not least in the USA, and a career in industrial science often seemed 'second best' to university or government service (Dennis, 1987; Swann, 1988). In Britain, technical education was rather '*infra dig*'. Oxford had finally taken science seriously, usually by importing talent from provincial universities, and big new cartels, like Imperial Chemical Industries, were respectable sources for research support and for employment of graduates – but still, too close an interest in technical matters was not good for an undergraduate's prestige. At Oxford, as a student or a fellow in chemistry, you were well advised to play down the 'stinks' and play up your musical talents (Morrell, 1997a).

Leading English physicians were well advised to keep abreast of medical research, but to subordinate it to the claims of experience and clinical judgement – to the *individuality* of both doctor and patient that was characteristic of biographical medicine (Lawrence, 1985; 1994). Specialists seemed to threaten this order of medicine, especially if supported by salaries rather than private practice. They suggested a world where the state might impose a 'division of labour' – each patient divided according to his diseases; each doctor specialising in particular conditions. (Analysis and rationalised production triumphing over biographical medicine and the liberal professions, as in the radiotherapy centres.) In most Western countries, supporters of the liberal order of medicine responded by stressing holism, natural history and humanities (Lawrence and Weisz, 1998; Timmermann, 2000).

Technosciences in and after the Second World War

This dynamic between the organisations of technoscience and the older, more independent forms of industry, medicine and university research was further transformed by the Second World War and its aftermath. The medium-term effects on the sciences were dramatic, especially in Britain and North America. The atom bomb is the obvious exemplar of a massive new project

with major consequences for the post-war world; it required a mobilisation of scientists and engineers that dwarfed the previous scale of *academic* enterprises.

Since before the First World War, physicists had guessed that nuclear reactions could produce explosions, and by the late 1930s some researchers were checking the possibilities on paper. Physicists in Britain and America, fearing that the Germans might be developing an atomic bomb, pressed for a research and development programme, and serious efforts began in the USA in June 1941. They involved huge new facilities on several sites, some of which depended on the proximity of major hydroelectric projects. They also involved massive relocations of technical staff, including immigrants from Europe, and the concentration of state-of-the-art equipment from many of the country's best physics laboratories. Some of the management was provided by universities, most by the US army, including its engineering and explosive divisions. In technical terms, the project was remarkably successful; the morality of the use of the atom bomb on Japan continues to be debated (Mendelsohn, 1997).

It seems reasonable to think of the bomb project as the origin of the third great technoscientific complex of the later twentieth century – *nuclear science and technology*. In the USA it included the commercial nuclear companies, the massive facilities for atomic physics used by physicists from universities, the new programmes in medical physics and radiotherapy, and the geneticists' surveys of radiation damage in Japan (Owens, 1997). In the UK, it centred on the publicly owned Atomic Energy Authority, with major installations around Oxford and in the north-west – openly developing civilian nuclear energy and radioisotopes, and covertly making plutonium for bombs. In these countries, and in the USSR, the nuclear complex was a major part of a general reconstitution of science and industry after the Second World War (Gowing, 1964; Kevles, 1987; Rhodes, 1986).

Many other aspects of science and industry were also reformed, chiefly by governments. The US military became by far the largest sponsor of research in American universities; military projects were deliberately 'spilled over' into peacetime initia-

tives, and they served as examples for future national campaigns – to reach the moon or to cure cancer (Studer and Chubin, 1980). University-industry-government complexes thrived in aviation, ballistics and radioastronomy, and then in space exploration (Geiger, 1992). In several countries, state support was crucial for the development of computing, for even in peacetime the military was the major sponsor and user of large-scale data processing. In Britain, wartime planning and science was carried forward in newly nationalised industries and services, including the National Coal Board and the National Health Service (Gowing, 1964; Price, 1976). The universities, and the high schools which fed them, benefited from greatly increased state support, especially for STM. Public scholarships were now available to all eighteen-year-olds with the grades requisite for university entrance; meritocracy was linked to faith in education, especially in science.

In Britain and especially in the USA, wartime production of penicillin led to the rapid development of new antibiotics, and of research-oriented pharmaceutical companies; from them came new drugs acting on the cardiovascular and nervous systems (Galambos and Sturchio, 1997; Goodman, 2000; Parascondola, 1980).⁷ The British National Health Service became a major funder of certain kinds of medical research, including salaried clinical professors in place of the part-time teachers who had relied on private practice. Clinical research became common in all teaching hospitals, often in association with pharmaceutical companies, and sometimes with the companies manufacturing medical equipment (Blume, 1992; 2000; Lawrence, 1997). In the USA, as we have seen, biomedical sciences benefited from the general increase in public funding for university research, and from the special research programmes instituted during the Korean war. All these developments strengthened the links between government agencies, pharmaceutical companies, universities and teaching hospitals.

Generally, in the UK as in the USA, the major STM programmes between 1945 and the 1970s could be described as *post-war* in a sense that is much more than merely chronologi-

cal. The major technoscientific complexes were formed or reshaped in the Second World War (and by the cold war and the Korean war), and in all these post-war technosciences, the state was central. That is obvious for Britain; it becomes obvious for the USA when the military investments are included.

Coda

In our history so far, the post-Second World War decades are easily seen as the product of major social forces put in play by industrialisation and strengthened by international competition and conflict from the end of the nineteenth century. That is how the post-war decades presented themselves – their ‘modernity’ was compounded from scientific principles, rationalised production, functional aesthetics, medical prowess, professional expertise and the welfare state. To defend the West, some of these capacities had to be devoted to the cold war, not least to nuclear deterrence – a clean, if risky, form of power. All the technoscientific networks were crucial to this form of civilisation and to this self-image – for providing electrical power systems and communications, for new pharmaceuticals and the conquest of disease, and for nuclear energy and the balance of power. All these complexes were built from the employment of scientists by companies and state agencies, and by linkages with universities. Partly by the use of new kinds of electronic computing, these networks commanded huge amounts of information and considerable analytical capacities, as well as major facilities for systematic invention and experiment.

We should not forget (as enthusiasts for ‘white-hot’ technology often did) that many companies succeeded more by finding new markets or cheaper processes than by technoscience, or that the industrial health of nations might depend more on competent labour management than on industrial research. We should remember (as exponents of social medicine then insisted) that public health depends more on healthy diets, good habits and sanitation, than on miracle cures. And we have Vietnam to remind us that military success could depend on public morale

rather than technical might. Indeed, it was partly such critical reactions which threw into relief the power of the 'military industrial complex' and its medical analogues.

Throughout the later twentieth century, the technoscientific complexes continued to grow, but from about the 1970s the politics of state and science changed considerably, perhaps especially in Britain. As we shall see in the next, and final, chapter, state support and direction declined *relative* to that of industry. Though the nuclear complex diminished by the end of the century, the electrical-electronic complex expanded around computers, and the pharmaceutical complex around molecular biology – but in both fields the directions came increasingly from commerce.

In Britain, by the end of the twentieth century, many public utilities had been denationalised and some state research agencies sold off, especially in agriculture. 'Business values' and professional management were being promoted in what remained of the public sector, including the health service and the universities. The government-funded research councils were being encouraged to think of themselves as part of the infrastructure of the national economy, and universities to focus on intellectual capital and the exploitation of discoveries. The rapid 'globalisation' of finance and business meant that a few international conglomerates dominated technological development worldwide. For these reasons, the end-of-the-century waves of technoscience – the informatics and the genetic engineering – were shaped by market values even more than would have been the case in the 1970s.⁸

It is in this context that 'public understanding of science' assumes a new importance. To those issues we turn in the next and final chapter.

Notes

- 1 On 'Big Science', which in my terms is mostly academic-led technoscience, see Capshew and Radar (1992); Galison and Hevly (1992); Krige (1997).